Next Generation Data Centers and Energy Management Systems

Lecture 6B:Chip Cooling: Heat Conduction and Thermal Interface Materials

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Department of Mechanical Engineering

Fall 2026

Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- **Chip Cooling: Heat Conduction and Thermal Interface Materials**
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Model heat conduction through a multi-layer chip cooling stack.
- Fundamentals of 2D transient heat conduction in isotropic materials.
- Modeling 2D transient heat conduction in heterogeneous materials.

Typical layout of a data center facility

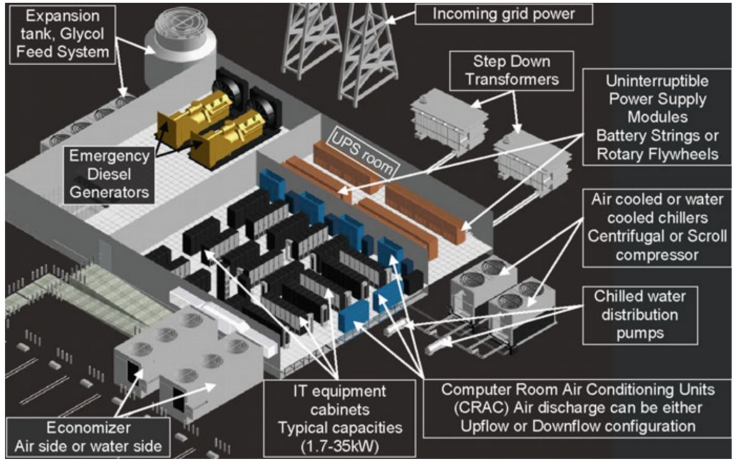


Figure 1: (Figure 1.1 in Ref. [1]).

Seminar 3

- Michael Donahue, Schneider Electric
- Date: October 8, 2026
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Heat Transfer Modeling: Conduction

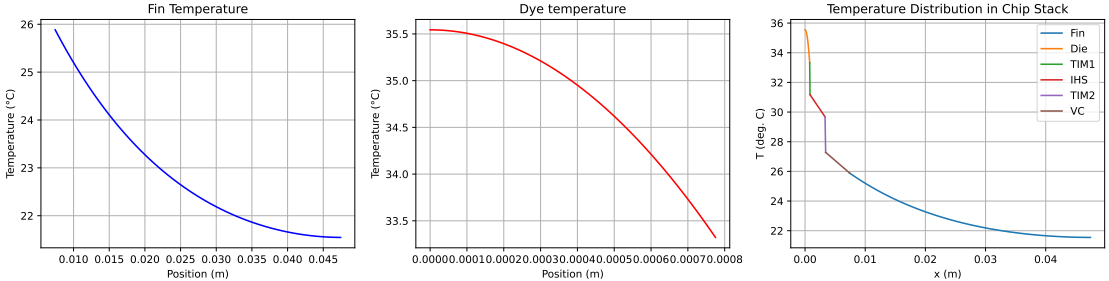
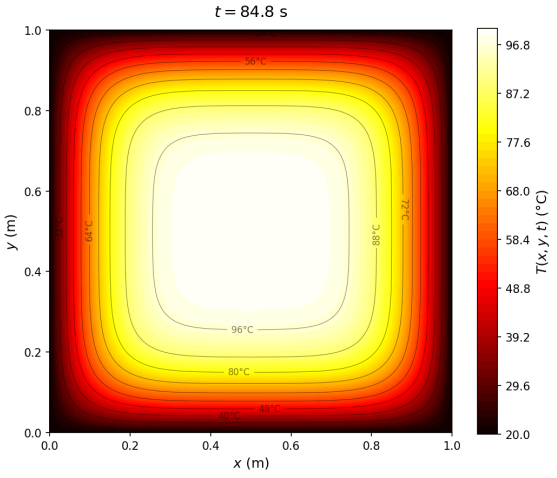
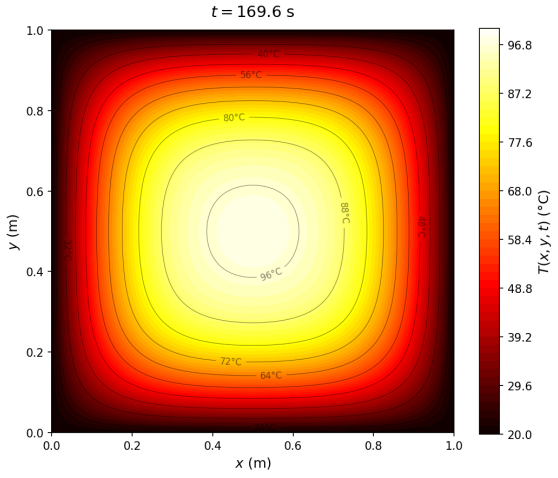


Figure 2: Temperature distribution in the chip stack.

Heat Transfer Modeling: 2D Transient Conduction, Analytical





Heat Transfer Modeling: 2D Transient Conduction, Numerical

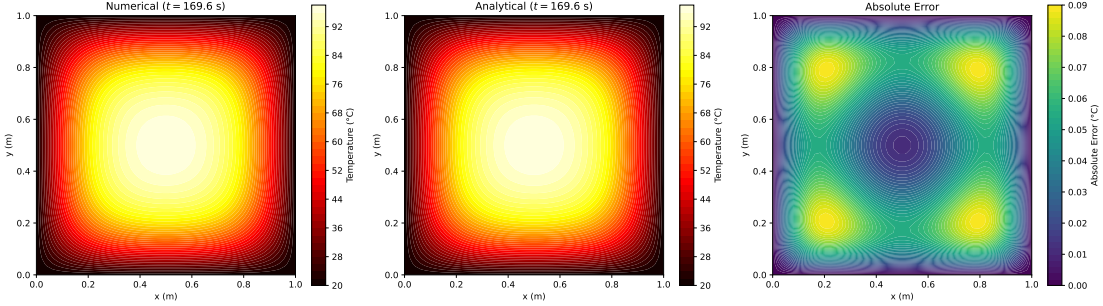


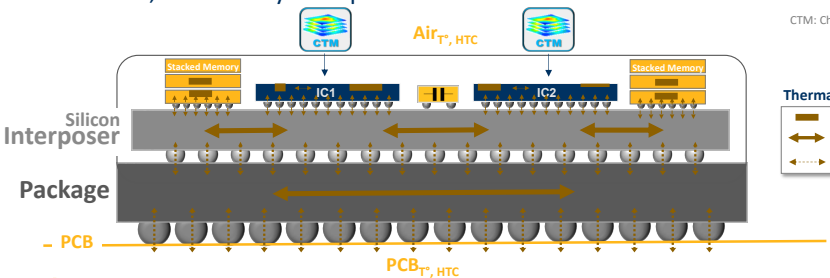
Figure 3: Analytical vs. numerical solution to the 2D transient heat conduction problem.

Chip Thermal Modeling for 3DICs

- Predict the thermal hotspot of the die across multiple on-die routing layers
- Hotspot locations can guide on-die sensor placement and thermal management
- Chip/chiplet level static and transient temperature response
- RedHawk-SC Electrothermal is our chip-level thermal sign-off solution, certified by multiple foundries



CTM: Chip Thermal Model



Thermal Conductive Paths

sources	✓ IC Power Density
intrinsic	✓ Metal Densities
TSV, Bumps	✓ Materials
	✓ Boundary Conditions

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Heat Transfer Modeling: 2D Steady Conduction, Heterogeneous Material, Numerical

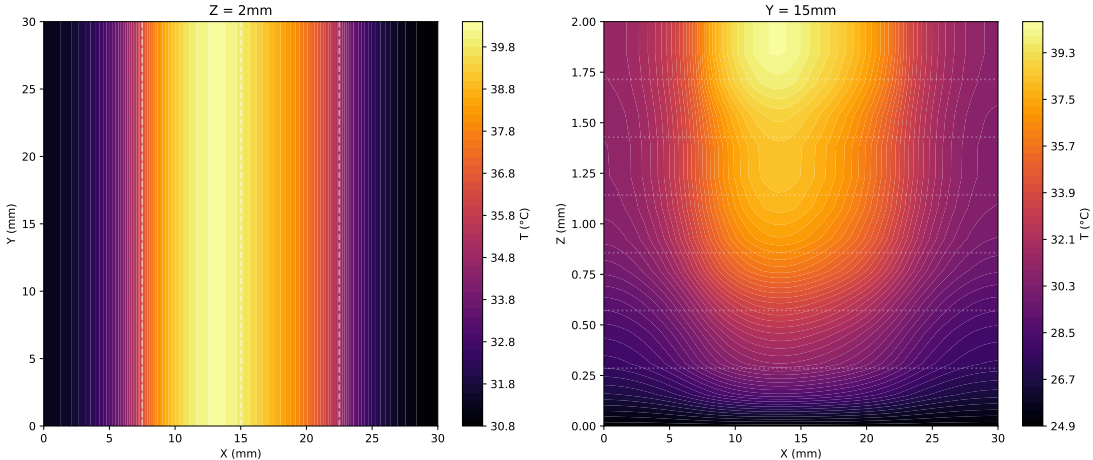


Figure 4: Temperature distribution in 3D IC.

References I

[1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.